

CPIS-351 Syllabus

Catalog Description

CPIS-351 IS Analysis and Architecture Design

Credit: 3 (Theory: 3, Lab: 2, Practical: 0)

Prerequisite: CPIS-250 , BUS-232

Classification: Department Required

The objective of this course is to introduce methods used in IS analysis in order to identify and characterize needs and to automate and create computer systems for them. The automated systems incorporate several technologies, and this course studies the way making optimum use of such systems for the users' service. The course emphasizes on the design phase activities and also presents design using structural and object-oriented techniques. Topics include system architectures design, traditional approach to design, object-oriented approach to design for applications, designing files and database, designing the user interface, designing the system interface and prototyping, controls and security, implementation, and support issues.

Class Schedule

Lab/Tutorial 90 minutes 1 times/week

Meet 50 minutes 3 times/week or 80 minutes 2 times/week

Textbook

John Satzinger, Robert Jackson, Stephen D. Burd, "Systems Analysis and Design in a Changing World", Cengage Learning; 5 edition (2008-03-28)

ISBN-13 9781423902287

ISBN-10 1423902289

Grade Distribution

Week	Assessment	Grade %
2	Homework Assignments 1	5
2	Graded Lab Work 1	2.5
5	Graded Lab Work 2	2.5
6	Quiz	5
7	Graded Lab Work 3	2.5
8	Exam 1	20
9	Homework Assignments 2	5
9	Graded Lab Work 4	2.5
10	Essay Writing	5
13	Group Project	20
14	Comprehensive Final Exam	30

Please note that the maximum accepted similarity rate for all submissions is 25% for Under Graduate Assignments. Please check with your instructor for more details and exceptions.

Last Articulated

April 6, 2021

Relationship to Student Outcomes

1	2	3	4	5	6
	X	X			

Course Learning Outcomes (CLO)

By completion of the course the students should be able to

1. Describe the difference between systems analysis and systems design (traditional and object oriented) (2)
2. Describe the major components, levels of design and design phase activities (traditional and object oriented) (2)
3. **Articulate design principles including separation of concerns, information hiding, coupling and cohesion, and encapsulation. (2)**
4. Identify popular software architectures (2)
5. Use CRC cards to define class responsibilities and collaborations (2)
6. **Create appropriate design models for the structure and behavior of software products from their requirements specifications. (2)**
7. Define class responsibilities and collaborations by applying Responsibility Assignment Patterns (2)
8. **Design a relational and object oriented database schema based on a class diagram (2)**
9. **Explain the importance and design of system interfaces, controls, and security (2)**
10. **Explain design patterns and recognize various specific patterns (2)**
11. **Communicate orally and in writing the results of applying the acquired knowledge of software engineering phases of analysis and design and their relevant techniques to a small sized group project. (3)**
12. Describe various types of software tests, versioning and explain how and why each is used (2)
13. List various approaches to data conversion and system installation and describe the advantages and disadvantages of each (2)

Coordinator(s)

Dr. Sahar Jambi, Assistant Professor

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Topics Coverage Durations

Topics	Weeks
Essentials of Design and the Design Activities (Ch-1)	2
The Traditional Approach to Design (Ch-2)	1
The Object-Oriented approach to Design: Use Case Realization (Ch-3 - Part-1)	5
GRASP patterns (Ch-3 - Part-2)	1
Designing System Interfaces, Controls, and Security (Ch-4)	1
Designing database and controls (Ch-5)	2